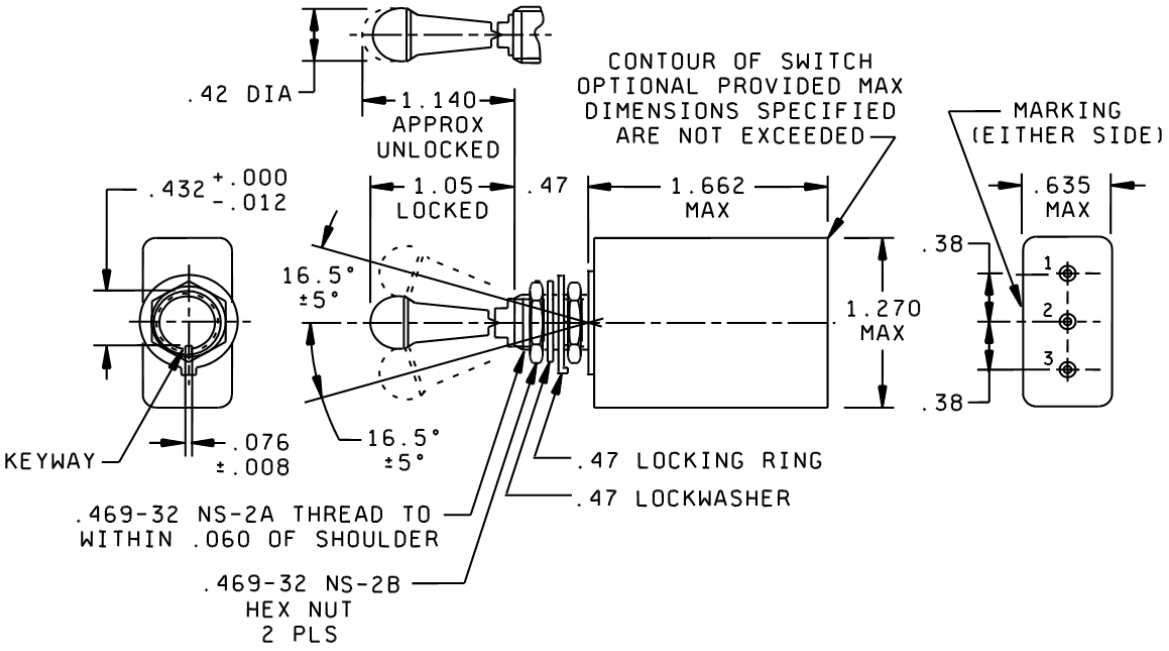
INCH-POUND
MS27781F
4 May 2024
SUPERSEDING
MS27781E
24 November 2021

DETAIL SPECIFICATION SHEET

SWITCH, TOGGLE, ONE POLE, ENVIRONMENTALLY SEALED, LEVER LOCK, INTEGRATED WIRE TERMINALS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-3950.



NOTES:

1. Dimensions are in inches
2. Unless otherwise specified, tolerance is ± .020 or two place decimals and ± .005 for three place decimals.
3. For hardware detail specifications see appendix of MIL-DTL-3950.
4. Contact installing/ removal tool in accordance with M81969/14-03 per SAE-AS81969/14.
5. Grommet sealing plug, in accordance with MS27488-16-2 per MS27488.
6. Terminal shall adequately accept a wire contact within dimensional limits of M39029/1-102 per SAE-AS39029/1.
7. Direction of internal mechanism movement is opposite to the direction of the toggle movement.

FIGURE1. Dimensions and configuration

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:51Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Maximum weight is .108 pound.

The force required to release the locking mechanism shall be 3 to 5 pounds.

Locking arrangement: Positive locking shall be accomplished and shall prevent motion of the toggle movement

The locking means at the top of the toggle bushing shall be capable of withstanding a torque of 20 inch-pounds applied in both directions immediately following the humidity test.

The sealing grommet shall seal on a smooth wire insulation of .068 to .103 diameters.

The terminal end of switch shall be color coded blue to indicate contact size.

Sealing plugs may be used in nonfunctional grommet holes.

Electrical rating: See table I.

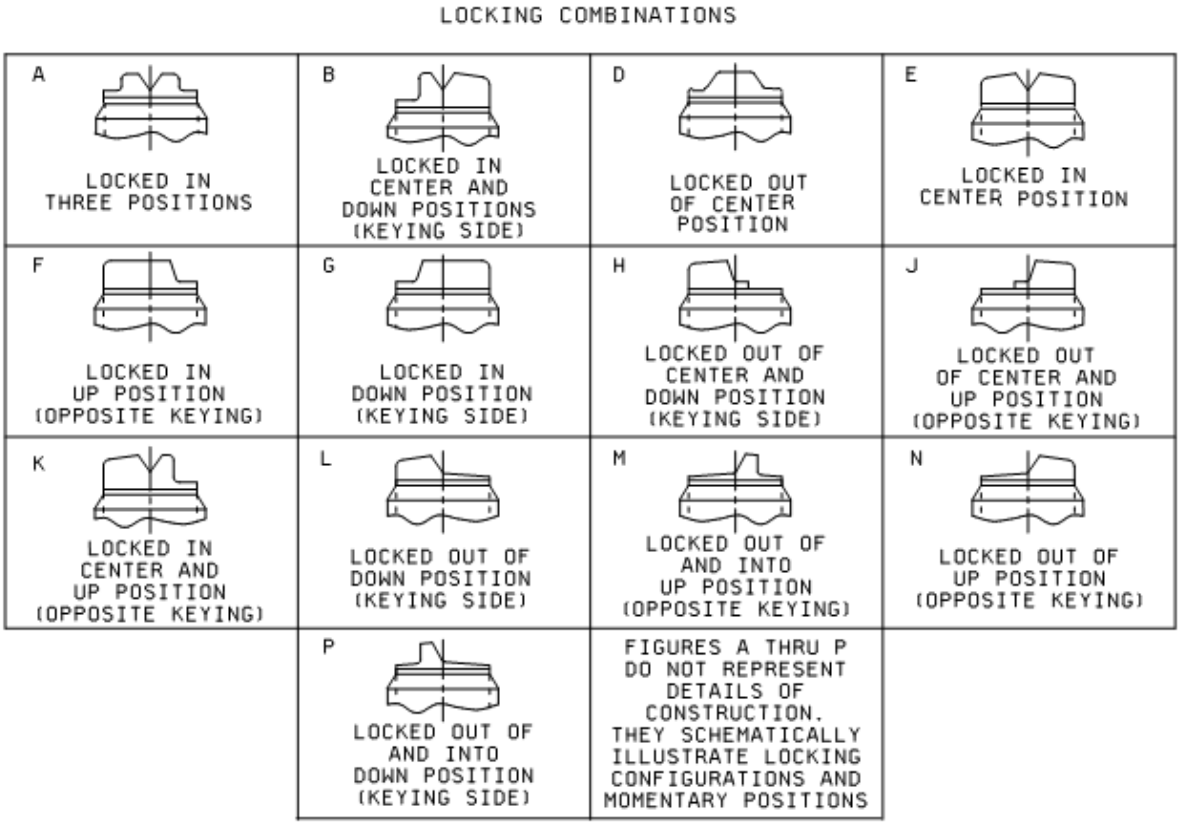


FIGURE 2. Locking Combinations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:51Z
Check the source to verify that this is the current version before use.

TABLE I: Detail Requirements

MS part No	Available locking combinations	Circuit with Toggle Lever In			Current capacity (amperes) 28 volts DC			Current capacity(amperes) 115 volts, 60 and 400 Hertz AC		
		Keying side	Center	Opposite keying side	Lamp-Load circuit	Resistive circuit	Inductive circuit	Lamp-Load circuit	Resistive circuit	Inductive circuit
MS27781-21	ALL	1-2 ON	OFF	2-3 ON						
MS27781-22	D, F, G	OFF	NONE	2-3 ON						
MS27781-23	D,F,G	1-2 ON	NONE	2-3 ON	5	20	15	3	15	10
MS27781-24	E,F,K,M	NONE	OFF	2-3 ON						
MS27781-25	F	NONE	MOM OFF	2-3 ON						
MS27781-26	F	1-2 MOM ON	NONE	2-3 ON						
MS27781-27	E,L,N	1-2 MOM ON	OFF	2-3 MOM ON						
MS27781-28	E	1-2 MOM ON	OFF	NONE	4	15	10	2	15	7
MS27781-29	F	MOM OFF	NONE	2-3 ON						
MS27781-30	F	1-2 MOM OFF	NONE	OFF						
MS27781-31	E,F,K,L,M,N	1-2 MOM ON	OFF	2-3 ON						
MS27781-32	E	NONE	ON 1-2	2-3 MOM ON						
MS27781-33	E,F,K,M	NONE	ON 1-2	2-3 ON	5	20	15	3	15	10

Test Requirements

a) During all tests. Switches shall be fully wired with appropriate wire and terminal contacts.
b) Contact voltage drop- the contact drop with two terminals and the switch contact in series shall not exceed 8 millivolts measured from one wire contact through the contacts to the other wire contacts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:51Z
Check the source to verify that this is the current version before use.

Referenced documents

MIL-DTL-3950	MS27488
SAE-AS39029/1	SAE-AS81969/14

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity
DLA – CC

(Project 5930-2024-019)

Review activities
Army – AR, AV, MI
Navy – EC, MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:51Z
Check the source to verify that this is the current version before use.